

Lean-checked results of team HTPeo at OpenMath 2026: a new upper bound for the K_4 Ramsey multiplicity constant, partial results on the Dvořák–Mohar–Šámal conjecture, and formalizations of known results

Woohyuk Kang* Hyunjin Lee† Sanghyeon Lee‡ Youchan Oh§

3 October 2026

Repository: <https://github.com/lavaskiller/openmath-2026-hypeo>, tag
submission-2026-10-03-v8

Abstract

We report what team HTPeo obtained during the OpenMath 2026 event (27 September – 3 October 2026), in which only formalized results count. (1) A weighted two-colouring template on 1024 blocks shows $c_4 \leq 200137750014677779172692777477/6640289795261907019148161833841 \approx 0.0301399120$ for the K_4 Ramsey multiplicity constant, below the previously best value $10486266368/768^4 \approx 0.0301422734$ (McKay); the whole chain from the template to the bound on c_4 is checked in Lean 4 with standard axioms only. (2) For the Dvořák–Mohar–Šámal conjecture (star chromatic index of subcubic multigraphs at most 6) we do *not* prove the conjecture; we prove in Lean star 5-edge-colourings of several infinite families, star 6-edge-colourability of all bridgeless cubic loopless multigraphs on at most 14 vertices, an equivalent form of the conjecture, and a reduction to named open hypotheses. (3) We formalize known results attached to 16 Erdős problems, stated exactly as in the formal-conjectures repository, and Schönberger’s and Petersen’s theorems for Mathlib’s `SimpleGraph` (17 families, 24 theorems). We also list our results on the competition’s scored tasks (“hills”), three of which carry Lean certificates, and we state plainly what is not checked, what was already known, and what the work cost. Most of the mathematics and all of the Lean code were produced by AI systems under human direction; the human review was limited (Section 7).

1 Setting

OpenMath 2026 (the handbook is titled *Open Problems Hack at MIT*) ran from a status freeze at noon Eastern on 27 September 2026 to the submission deadline at 00:00 EDT on 3 October. Problems are posed as *hills* on the AutoLab platform: versioned tasks with an evaluator and a leaderboard. In the handbook’s words a passing hill “is not itself a mathematical proof”; formal checking, statement fidelity, literature status, attribution and review are separate gates. Our Ramsey and DMS entries are proposed under mode M3A (open problems outside the curated focus set), the formalizations under M2 (known mathematics, counted in families).

All Lean results below use Lean 4.33.1 and Mathlib v4.33.1 (the Erdős files: Mathlib as pinned by `google-deepmind/formal-conjectures` at commit `df3f12d7`). Every claimed theorem compiles with exit code 0 and `#print axioms` lists only `propext`, `Classical.choice` and `Quot.sound`. No claimed theorem depends on `sorry`, an added axiom, or `native_decide`. The

*HTPeo, Kyung Hee University, CS&E. woohyuk@khu.ac.kr

†University of Cambridge. hl728@cam.ac.uk

‡Korea University, Security. ymhlsh4065@korea.ac.kr

§Seoul National University, TI. thomasoh0408@snu.ac.kr

repository contains the sources, the build logs with the axiom output, checksums, and the exact commands.

2 The K_4 Ramsey multiplicity constant

Let $\text{minMono}_{K_4}(m)$ be the least number of monochromatic K_4 over all red/blue colourings of the edges of K_m . The constant $c_4 = \lim_m \text{minMono}_{K_4}(m)/\binom{m}{4}$ exists. Thomason disproved Erdős' conjecture $c_4 = 1/32$; the best published upper bound we are aware of is McKay's $10486266368/768^4 \approx 0.0301422734$, announced in the closing note of Parczyk, Pokutta, Spiegel and Szabó (arXiv:2206.04036v3), obtained by local search from a 768-vertex Cayley graph; the best lower bound cited there is 0.0296 (Grzesik, Lee, Lidický, Volec). McKay's value is the reference of the AutoLab hill `clique-cluster-ramsey-multiplicity`.

Result. A symmetric weighted template on 1024 blocks (integer weights, total $Q = 50762939$, each pair of blocks and each block's interior coloured red or blue) has monochromatic- K_4 density

$$\frac{N}{Q^4} = \frac{200137750014677779172692777477}{6640289795261907019148161833841} \approx 0.030139911990,$$

i.e. hill metric $\lceil 10^{12} \cdot N/Q^4 \rceil = 30,139,911,990$ against the reference 30,142,273,432. Blowing the template up gives colourings of K_m for all large m with at most that density, hence

```
theorem ramseyMultK4_limit_lt_ref :
  ∃ L : ℝ, Filter.Tendsto (fun m : ℕ => (minMonoK4 m : ℝ) / (m.choose 4 : ℝ)) Filter.atTop (nhds
    L) ∧
    L < 10486266368 / 768 ^ 4
```

together with `ramseyMultK4_le_sol` (the limit is at most the fraction above), `sol_density`, `sol_ppt`, `sol_symm` and `minMonoK4_density_le_sol` (every $m \geq 4$ satisfies the bound).

How the proof is checked. The finite computation (the number of monochromatic K_4 in the weighted template) is split into 2×1024 per-block kernel evaluations (`decide +kernel`) on a packed bit-mask representation whose agreement with the literal specification is proved once for all templates; the blow-up lemma, the existence of the limit (vertex-deletion averaging) and the comparison with the reference are ordinary Lean proofs. The certificate has 2048 generated chunk modules; it built with two processes in about two hours (3.9 CPU hours, at most 3.3 GB per process). The transcription of the template from JSON to Lean is done by a script and checked by a second script; it is not itself proved.

How the template was found. Starting from the hill's 768-vertex Cayley seed: simulated annealing and tabu search on the colour orbits under the automorphism group, splitting of near-twin blocks up to 1024 blocks, continuous weight optimisation with rounding to integers, and basin hopping; every candidate was re-scored exactly with the hill's own evaluator. A first template (density_ppt 30,139,933,996, 2 October) was improved on 3 October and its certificate regenerated. On the hill's validation board the result is first of 13 accounts at the time of writing, 224,043 ppt ahead of the second. It has no final-mode (held-out) evaluation; one attempt failed for a technical reason. The improvement is numerical (about 2.4×10^{-6} in c_4) and does not determine c_4 .

3 The Dvořák–Mohar–Šámal conjecture

The conjecture states that every subcubic multigraph has star chromatic index at most 6 (a star edge-colouring is proper and has no bichromatic path or cycle of length four); the best

published general bound we know is 7. **We do not prove the conjecture.** Proved in Lean (`entries/dms-star6`):

- star 5-edge-colourings of the flower snarks J_n (n odd, $n \geq 5$), the Goldberg snarks, the generalized Petersen graphs $GP(n, k)$ for all $1 \leq k \leq 15$ and $n \geq 2k + 1$ (except $GP(3, 1)$), the Möbius ladders and Petersen-type inflations, each as a statement for the whole infinite family;
- star 6-edge-colourability of every bridgeless cubic loopless multigraph on at most 14 vertices;
- an equivalent form `dms_iff_cubic16` reducing the conjecture to bridgeless cubic multigraphs on at least 16 vertices together with a condition on their leaf sets;
- a chain “named open hypotheses $\Rightarrow$ conjecture”; each hypothesis is open.

The graph families are given as explicit edge lists; their agreement with the textbook definitions is checked by a script, not proved as an isomorphism. These results came out of an automated multi-agent “harness” that kept a graph of verified facts and cross-checked them with a second model family; the informal studies of the remaining core case (cyclically 4-edge-connected cubic graphs) are in the archive and are not Lean-checked.

4 Formalizations of known results (M2)

Each Erdős file is the pinned formal-conjectures file with the `sorry` of the target theorem replaced by a proof; the statements are textually identical to formal-conjectures. “New” means only that searches of formal-conjectures, the TheJustinSunPrize/awards pull requests, plby/lean-proofs and other public collections found no earlier formal proof; several candidates were dropped as duplicates.

Family	Lean theorem(s)	Tier
G-PM	<code>simple_schoenberger</code> , <code>simple_petersen_connected</code> (Schönberger, Petersen, connected case)	substantive
E942	<code>erdos_942.variants.limsup</code>	substantive
E44	<code>greedy_sidon_construction</code>	substantive
E123	<code>erdos_123.variants.powers_2_3</code>	substantive
E918	three variants <code>eq_aleph_0...</code>	substantive
E292	<code>variants.mul</code> , <code>two_mul</code> , <code>prime_pow</code>	substantive
E395	<code>erdos_395.variants.one</code> (Carnielli–Carolino counterexample)	substantive
E698	<code>erdos_698.variants.erdos_szekeres</code>	substantive
E939	<code>erdos_939.variants.seven</code> , <code>eight</code>	substantive
E477	<code>variants.S_sq</code> , <code>degree_two_dvd_condition_b_ne_zero</code> (Sekanina; quadratic case)	substantive
E295, E703, E748, E1136	<code>helper lemma</code> , <code>variants.zero</code> , <code>lower_bound</code> , <code>multiples_of_three</code>	minor
E358, E619, E1148	<code>f_id</code> (odd divisors), <code>add_edges_diam_three</code> , <code>variants.weaker</code>	minor

The four families E477, E358, E619 and E1148 were added on the last day. Erdős problem #1038 is treated separately (Section 6).

5 Results on the competition hills

Leaderboard standings were read on 3 October 2026 at 02:44 UTC; accounts with identical metrics share a rank.

Hill (board)	Result	Standing	Lean
K_4 Ramsey multiplicity	density_ppt 30,139,911,990 (Section 2)	1st of 13	yes
Kobon triangles ($n = 39$)	471 triangles with 39 lines	1st of 3	yes
Busy Beaver 6	machine 0LE1LA_1LC1RH_1RA1RF_ORFORD_1RD1LB_1LC1RD 249,881 steps, 554 ones, span 735	tied 1st, 3 of 12	yes
Grothendieck witnesses	CHSH matrix, ratio 1393/985, gap_ppm 1,414,213, 80 bits	tied 1st, 7 of 9	yes
Kobon triangles ($n = 18$)	93 triangles	tied 1st, 12 of 15	no
3×3 matrix multiplication	rank 23, support 139 (known scheme)	tied 2nd, 7 of 11	no
Collatz modular descent	234 rules, full coverage	5th of 8	no
Ramsey, held-out board	density_ppt 30,141,921,123	only entry	no

Kobon triangles, 39 lines. Starting from the Füredi–Palásti arrangement (468 triangles), simulated annealing on the line parameters found 39 lines with integer coefficients bounding 471 triangles. The Lean certificate (`Kobon.kobon_nonoverlapping`) states that there are 39 pairwise distinct lines and 471 index triples, each bounding a non-degenerate triangle whose open interior meets none of the lines, the triangles being pairwise disjoint; `Kobon.certificate` checks the list against the signed hill report. The upper bound is $\lfloor 39 \cdot 37/3 \rfloor = 481$. Whether 471 exceeds every published value for 39 lines was checked only by an AI helper (OEIS A006066 and one public gallery), **not by a human**; we do not claim it as new.

Busy Beaver 6. The Lean project (core Lean, no Mathlib) proves that the machine above halts after exactly 249,881 steps, reaches all six states, and leaves 554 ones on a span of 735 cells, and that it is accepted by the hill exactly when the budget is at least 249,881. It reproduces the board’s best value. That 249,881 is the largest halting time below the hidden budget is *not* proved; a separate computation excluded 31 of 32 structural classes of machines and was stopped before the last class was finished.

Grothendieck constant. For the CHSH matrix with rational unit vectors the certificate proves that every constant K satisfying the finite-dimensional Grothendieck inequality has $K \geq 1393/985$, and with exact vectors $K \geq \sqrt{2}$; Tsirelson’s bound for this matrix shows that no witness on it can score higher. All of this is classical.

6 Erdős problem #1038

Two team members produced a proof (with Claude, cross-checked with GPT) that the infimum of the measure of $\{x : |f(x)| < 1\}$ over monic real polynomials with all roots in $[-1, 1]$ equals an explicit constant $D \approx 1.8344304757$, and that the supremum is $2\sqrt{2}$, and formalized it in Lean 4.34.1 (315 modules). During the work they found that a proposed solution with the same approach had been published and that a Lean proof was already public (S. Wang; imported into `plby/lean-proofs` on 15 September 2026). The team’s development is a reconstruction with a different implementation. **We do not claim it.** The package and our partial rebuild with Lean 4.33.1 (supremum and classical lower bound re-checked; the exact infimum not, for memory reasons) are in `entries/erdos-1038`.

7 Provenance, review and cost

The DMS results came from an automated harness on a 16-core server: Claude agents proposed and proved lemmas, a Lean verification service accepted facts, and GPT models audited them; about 786 informal facts were recorded. The Ramsey search, the certificates, the Erdős formalizations and the hill work were done in Claude Code and Codex sessions directed by team members. Recorded

AI usage of the team lead’s server and laptop was about 8.1×10^7 output tokens in 2,145 sessions, all under subscription plans; the other members’ usage was not recorded as numbers. Human review: the team lead read the summaries, claimed statements and statement-correspondence notes of the three packets; this was not a line-by-line check of the proofs, and one member declared that no human review was done on their part. All proofs were written by AI models and checked by the Lean kernel and by scripts.

Limitations

- The Ramsey bound improves the constant numerically only; c_4 is not determined.
- The DMS conjecture is not proved; every hypothesis of our reduction is open.
- The M2 results are known mathematics; “first formalization” rests on searches with stated limits.
- The hill evaluators themselves are not formalized; where we compare with them, we compare against a hand transcription or against the signed report.
- Literature statements marked as unchecked by a human have not been checked by a human.

Availability

Everything is in <https://github.com/lavaskiller/openmath-2026-hypeo> at tag `submission-2026-10-03-v` per entry a packet (`PACKET.md`), the Lean sources with build and axiom logs, `SHA256SUMS`, and the exact reproduction commands; an archive with a timeline, statistics and findings. The repository was private to the team until the deadline.